

Amperometric Immunosensor for Chagas' Disease Using Gold CD-R Transducer

Marcos Vinicius Foguel, Glauco Pilon dos Santos, Antonio Aparecido Pupim Ferreira, Hideko Yamanaka, Assis Vicente Benedetti*

UNESP – Univ Estadual Paulista, Institute of Chemistry, Rua Prof. Francisco Degni 55, 14800-900, Araraquara, SP, Brazil
*e-mail: benedeti@iq.unesp.br

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Abstract

Chagas' disease has been spread mainly through blood transfusion or blood components, therefore the development of a device for neglected diseases with high specificity and sensitivity is important to public health. This work describes the development of an amperometric immunosensor for determination of Chagas' disease through a gold-based electrode obtained from a recordable compact disc modified with 4-(methylmercapto)benzaldehyde for the immobilization of Tc85 protein of the *Trypanosoma cruzi*. The immunoassays were carried out using positive and negative sera from Chagas' disease patients and immunoglobulin conjugated with peroxidase enzyme. The immunosensor presented $-0.949 \mu\text{A}$ as cut-off value and was applied in sera samples.

Keywords: Chagas' disease, Gold CDtrodes, Immobilization, Self-assembled monolayer, Sensors

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1 Introduction

Chagas' disease is a parasitic infection caused by the hemoflagellate *Trypanosoma cruzi* (*T. cruzi*) [1]. According to World Health Organization 10 million people are infected currently, mostly in Latin America, but it has now spread to other continents [2,3]. The natural transmission of this disease is through the bite of contaminated hematophagous insects, which act as vector. However, there are other nonvector transmissions, as via blood products, organ transplant, and oral transmission through ingestion of food contaminated with the parasites [4,5].

There are standard techniques to detect antibodies present in contaminated patients with Chagas' disease, as indirect hemmagglutination (IHA), indirect immunofluorescence (IIF) and enzyme-linked immunosorbent assay (ELISA) [6]. However, a low cost device for such neglected diseases should still be provided. Therefore, the development of immunosensor presents an advantageous technique due to its sensibility, specificity, stability and ease of handling. In addition, it requires small sample volumes, thus lowering costs when compared to the conventional analytical methods [7–9]. Belluzo et al. [10] developed an immunosensor to perform indirect immunoassays with amperometric detection using tailor-made chimeric receptors to react with the analyte, specific anti-*Trypanosoma cruzi* immunoglobulin G (IgG) using thiol-modified gold electrodes. Diniz et al. [7] developed a biosensor for Chagas' disease, based on impedance methodology, immobilizing two different recombinant antigens (cytoplas-

mic repetitive antigen and flagellar repetitive antigen) on an oxide layer of a gold electrode.

An immunosensor is a sensory device that contains antigens or antibodies bound to the surface of a transducer, which converts the biological signal into a measurable electrical signal [11]. These molecules must be immobilized on an electrode and maintain its biological activity on the transducer [12]. A procedure very useful for this is the formation of a self-assembled monolayer (SAM) on the electrode surface [13]. SAM consists of the interaction of a highly organized organic molecule layer on the electrode surface where one of the functional groups of the selected organic molecule has the function of providing the biological materials immobilization (e.g. enzymes, proteins, nucleic acid, antibody, etc) via the free functional group of SAM [14]. Organic monolayers with thiol or disulfide groups on the electrode surface are of great interest due to the sulfur binding strongly to the gold surface [15]. A molecule that presents this property is 4-(methylmercapto)benzaldehyde (SBZA), which has a mercapto group and maintains a free aldehydic group [16].

As the SAM formed by molecules containing thiol groups have strong affinity for gold due to the high S-Au bond energy [15,17], transducers of gold are widely used in biosensor development. Metallic bases prepared by sputtering, can be substituted, with advantages, by metallic substrates obtained from recordable compact discs (CD-Rs) [18]. Due to the simplicity of the construction and low cost, these devices can be used and, if necessary, discarded in cases of fouling, surface oxide formation, ir-

reversible adsorption, etc [19]. These electrodes (CDtrodes) have shown an electrochemical performance comparable to commercial gold electrodes with superior versatility and very low cost. Also, they are user-friendly because there is no necessary for electrode polishing [20].

In the literature, there are some reports on the use of CDtrode for different purposes, such as the chronoamperometric determination of Ce(IV) in acidic aqueous solutions in order to study the kinetics and the mechanism of polymerization reactions with Ce(IV) as the initiator [21], determination of selenium(IV) in batch system [22] and employed as support for electrochemical synthesis of bimetallic Pt and Au nanoparticles by means of SAM [23]. The use of CDtrodes in the development of biosensors was also reported as amperometric biosensor for the detection of Hepatitis C virus through the reaction of hybridization of DNA immobilized on the electrode surface with the complementary DNA from serum samples of patients [24], and evaluation of the behaviour of peroxidase enzyme immobilized on gold CDtrodes modified with SAM of 3-mercaptopropionic acid (MPA). In this case, the streptavidin molecules were linked to MPA and then the biotin-HRP conjugated was immobilized [25].

In this work, the development of an amperometric immunosensor for detection of anti-*T. cruzi* antibody using a gold CDtrodes transducer modified with SAM of SBZA was reported employing electrochemical measurements as linear sweep voltammetry (LSV), cyclic voltammetry (CV) and chronoamperometry. The immunosensor was applied to investigate serum of patients for Chagas' disease.

2 Experimental

2.1 Reagents and Solutions

The construction and cleaning of CDtrodes were, respectively, carried out with 69–70% HNO_3 and 95–98% H_2SO_4 (both were purchased from J. T. Baker). The SAM was formed by using 95% 4-(methylmercapto)benzaldehyde ($\text{C}_8\text{H}_8\text{OS}$ – Sigma-Aldrich). The *T. cruzi* Tc85 protein [26] (kindly provided by W. Colli and M. J. M. Alves, of the Institute of Chemistry, São Paulo University – USP, SP, Brazil) was diluted in 0.05 mol L^{-1} phosphate buffer solution at pH 7.5. Positive and negative sera of Chagas' disease were obtained from Diagnosis Laboratory of Faculty of Pharmaceutical Sciences (Unesp, Araraquara, SP, Brazil). The blocking solution was prepared with bovine serum albumin (BSA) in 0.02% salt phosphate buffer solution-Tween (PBS-T) and the anti-human immunoglobulin (IgG)-peroxidase (Sigma-Aldrich) was diluted in BSA solution.

The electrochemical measurements were performed in 0.1 mol L^{-1} phosphate buffer solution at pH 7.0, containing 1×10^{-3} mol L^{-1} H_2O_2 (Merck) and 3×10^{-3} mol L^{-1} KI (Aldrich) or $\text{Fe}(\text{CN})_6^{3-/4-}$ (Sigma-Aldrich).

All solutions were prepared with ultrapure water of Milli-Q quality (resistance $18.2 \geq \text{M}\Omega \text{ cm}$, Millipore Inc. USA).

2.2 Apparatus and Electrochemical Cell

The chronoamperometry, CV and LSV measurements were carried out using a potentiostat/galvanostat AUTO-LAB PGSTAT 302. Energy Dispersive X-ray spectroscopy (EDXS) was obtained with a JEOL JSM 7500F scanning electron microscope. Atomic force microscopy (AFM) images (tapping mode) were obtained using Agilent Technologies 5500 Scanning Probe Microscope (Agilent Technologies, Inc., California Street, USA). P-doped Si cantilevers with a force constant of 20–80 N m^{-1} and resonance frequency of 239–341 kHz were purchased from Nanosensors.

For electrochemical measurements a conventional electrochemical cell with a volume of 5 mL of supporting electrolyte and a three electrode system were used: gold CDtrode ($A_{\text{geom}} = 0.071 \text{ cm}^2$ and active area of 0.091 cm^2 by Randles-Sevcik equation [27]), $\text{Ag}|\text{AgCl}|\text{KCl}_{(\text{sat})}$ and platinum wire ($A_{\text{geom}} = 4 \text{ cm}^2$) as working, reference and auxiliary electrodes, respectively.

2.3 Construction of Gold Electrodes (CDtrodes)

The working electrode was constructed from a recordable compact disc (Mitsui Archive Gold CD-R 100 which after surface preparing were named CDtrodes) containing a gold film with thickness between 50 and 100 nm. To access the metal layer of the CD-R, concentrated HNO_3 was added on the surface, after 5–10 min the protective layers were totally removed and the gold surface was washed thoroughly with distilled water [18]. The CD-R was cut and the working area of the electrode was pre-set with perforated galvanoplastic tape. For the electrical contact of the CDtrodes, a laminated copper wire was fixed and insulated with polytetrafluoroethylene (PTFE) so that the wire does not come into contact with the electrolyte solution, thus avoiding the oxidation and reduction of this copper [28]. All CDtrodes are disposable, so they were used only once.

2.4 Preparation of Immunosensor

Figure 1A shows the surface morphology of the bare CDtrode, containing a gold layer approximately 140 nm thick and constituted by peaks and vales. The steps of the gold surface modification and immunoassay are presented in Figure 1B.

The first step in the development of the amperometric immunosensor was the addition of 2×10^{-2} mol L^{-1} SBZA prepared in ethanol, for 30 min of incubation time in a moist chamber at 4°C , to slow the evaporation of ethanol, on gold CDtrode for the formation of SAM. The use of SBZA instead of cysteamine and glutaraldehyde [8] diminishes one step in the SAM formation process. After

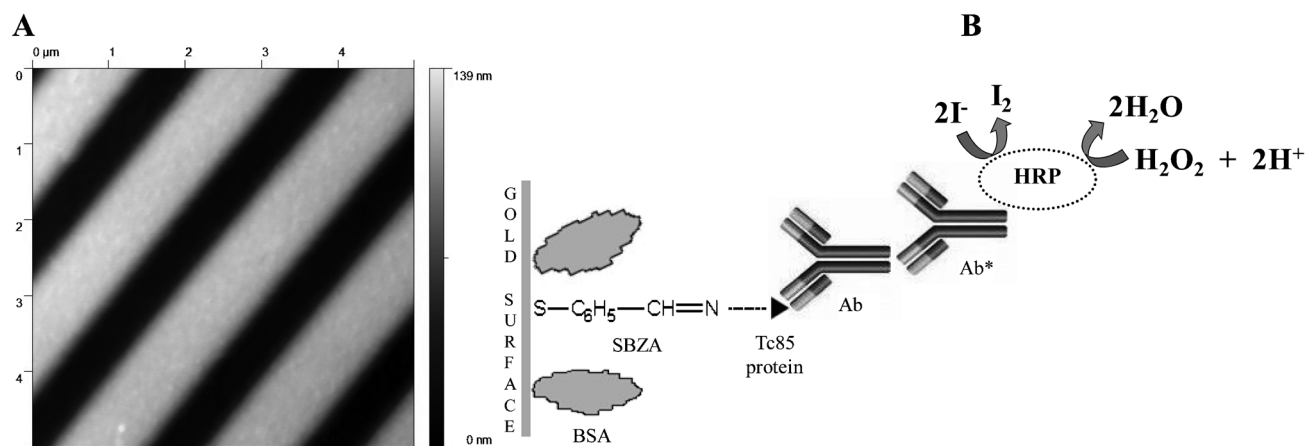


Fig. 1. AFM images of bare CDtrode surface (A) and scheme of immunosensor construction (B).

the incubation time, the washing of the electrode was performed by immersion in ethanol, under agitation for 3 times of 10 seconds. The antigen (Ag) Tc85 protein of the *T. cruzi* was immobilized on the SBZA modified surface by 6 h incubations at 4 °C. This condition was chosen after a detailed study about the influence of the incubation time of the Tc85 protein on SAM-modified screen printed gold-based electrodes (SPE) [29], in which the influence of the incubation time of the protein was clearly demonstrated by electrochemical impedance spectroscopy.

After antigen immobilization on the CDtrode surface, the immunoassay was performed using the following procedure (Figure 1): a) blocking of the sensor surface with 50 μL of PBS-T buffer solution containing 0.5% bovine serum albumin and incubation for 40 min at room temperature; b) washing in Milli-Q water; c) application of 50 μL of *T. cruzi* positive serum (Ab) diluted in blocking solution and incubation for 15 min; d) washing in Milli-Q water; e) application of 50 μL of the anti-human IgG antibody conjugated to peroxidase (Ab*) diluted in blocking solution and incubation for 15 min; f) washing in Milli-Q water; g) monitoring immunoenzymatic reaction by amperometric technique in hydrogen peroxide (substrate) and potassium iodide (mediator) solutions in the proportion of 1:3 [30] respectively, prepared in 0.1 mol L^{-1} phosphate buffer solution at pH 7.0 and incubation for 10 min.

After optimization of all experimental parameters the proposed device was applied for analysis of negative serum samples for Chagas' disease in order to determine the reactivity threshold. Finally the immunosensor was applied for analysis of positive and negative serum samples, previously analyzed using conventional techniques (ELISA, indirect immunofluorescence and indirect hemagglutination) [6].

2.5 Electrochemical Measurements

The CDtrodes were submitted to CV pretreatment of 10 cycles in 0.5 mol L^{-1} sulfuric acid solution, in the potential range between 0.0 and +1.5 V at 100 mV s^{-1} in order

to clean the surface. The reproducibility of modified electrodes with SAM and antibody was evaluated by cyclic voltammetry in the potential range of -0.4 to $+0.7$ V in 1×10^{-3} mol L^{-1} $\text{Fe}(\text{CN})_6^{3-/4-}$ at 50 mV s^{-1} .

To define the potential for amperometric monitoring, the linear sweep voltammetry was scanned from $+0.75$ to -0.9 V in 0.1 mol L^{-1} phosphate buffer solution at pH 7.0, in the presence and absence of 1×10^{-4} mol L^{-1} H_2O_2 and 3×10^{-4} mol L^{-1} KI at 50 mV s^{-1} .

The optimization and application of the immunosensor in positive and negative sera for Chagas' disease were carried out using electrochemical technique at controlled potential (chronoamperometry). A potential of -0.400 V was applied during 60 s in 0.1 mol L^{-1} phosphate buffer solution at pH 7.0, containing substrate and mediator. The current values of amperometric measurements were established at 40 s.

3 Results and Discussion

The gold CD-R is composed of polycarbonate plastic, photosensitive dye, thin layer of gold and finally a layer of lacquer to protect the metallic layer. After removal of the protective film by nitric acid the exposed CD-R is composed basically by gold as shown by EDXS (Figure 2); probably the carbon is providing by the dye film.

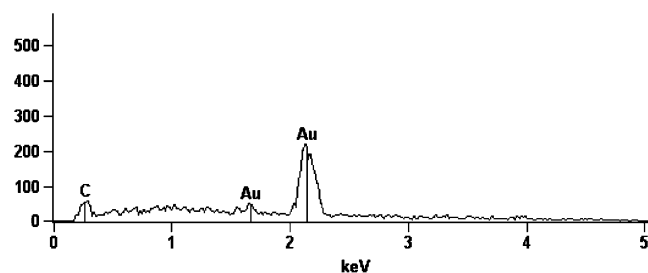


Fig. 2. EDX spectra for Au-CDtrode after removal of the protective film by HNO_3 .

3.1 SBZA Optimization to Form the SAM

The influence of incubation time on the formation of SAM ($C_{\text{SBZA}} = 4 \times 10^{-2} \text{ mol L}^{-1}$) on the gold surface was investigated by fixing 15 and 30 min. The modification efficiency was evaluated by calculation of the coverage degree of the electrode surface by mean of Equation 1 [31], where $I_{\text{p,SAM}}$ and $I_{\text{p,Bare}}$ are the reduction peak of ions $\text{Fe}(\text{CN})_6^{3-}$ for the modified and unmodified CDtrode, respectively.

$$\theta_{\text{CV}} = 1 - (I_{\text{p,SAM}} / I_{\text{p,Bare}}) \quad (1)$$

Thus it was calculated that after incubation for 15 min approximately 23% of the area of the electrode was covered by the monolayer, whereas after incubation for 30 min the estimated area of coverage increased to 49% (Figure 3). The investigation on SBZA concentration and incubation time to modify the gold CDtrode surface indicated that $2 \times 10^{-2} \text{ mol L}^{-1}$ during 30 min covered around 53% of the surface. Longer incubation times were not evaluated due to the rapid evaporation of the ethanol used to dissolve the SBZA, even employing a moist chamber at 4°C for this incubation.

Ferreira et al. [8] reported an amperometric immunosensor for Chagas' disease using a screen printed electrode (SPE) modified with cysteamine and glutaraldehyde for the formation of SAMs, however these modifications last six times longer than the SBZA modification. In addition the use of gold CDtrodes as electrochemical transducers reduces significantly the cost of the immunosensor, since the proposed method is around twenty times cheaper than the same device constructed with SPE.

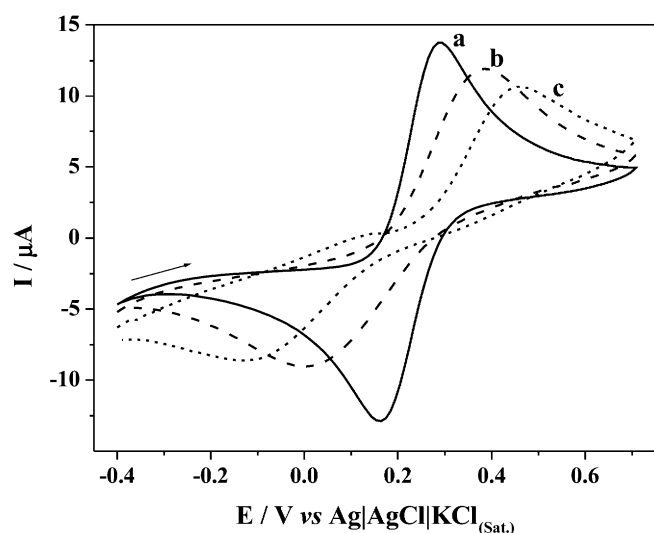


Fig. 3. Cyclic voltammograms in $1 \times 10^{-3} \text{ mol L}^{-1} \text{ Fe}(\text{CN})_6^{3-/4-}$ solution of unmodified (a) and modified gold CDtrodes with $4 \times 10^{-2} \text{ mol L}^{-1}$ SBZA with 15 (b) and 30 (c) min of incubation E/V vs. $\text{Ag}|\text{AgCl}|\text{KCl}_{(\text{sat})}$; $\nu = 50 \text{ mV s}^{-1}$.

3.2 Incubation Time of Tc85 Protein

The incubation times for immobilization of protein Tc85 on the gold surface modified with SBZA studied were 4, 6, 8 and 12 h (Figure 4). It can be concluded that the best time was 6 h, because this incubation time presented the lowest anodic peak current. With 4 hours of incubation, the protein is not immobilized on the SBZA monolayer yet, since the value of the anodic peak current is close to the value of CDtrode covered only with SAM. On the other hand for 8 and 12 h of incubation, probably a conformational change of the protein structure occurred, leading, for instance, to a more open structure since the anodic current peak increased again.

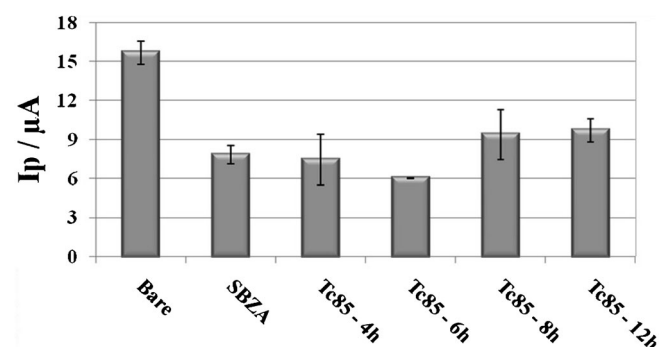
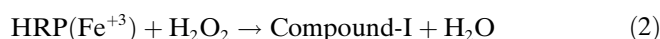


Fig. 4. Comparing bar graph of Au-CDtrode (bare), modified with SAM (SBZA) and with Tc85 protein immobilized on SBZA with different incubation times.

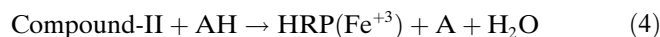
3.3 Amperometric Monitoring of Enzymatic Reaction

The definition of the potential to be applied to the amperometric monitoring of the enzymatic reaction, described below [30], was carried out by LSV.

The enzymatic reaction of peroxidase (HRP) can generically be represented by a sequence of three different steps, as shown by Reactions 2–4 [32–34]. The enzyme catalyzes the reduction of hydrogen peroxide oxidizing the hemi group (Equation 2):

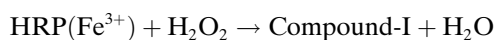


The oxidized enzyme (Compound-I) is reduced to Compound-II (Equation 3) which is further reduced to the $\text{HRP}(\text{Fe}^{+3})$ active form (Equation 4):



Reducing agents (AH) like iodide, phenols and others are able to donate electrons to both compounds I and II. Consequently, the reduction of species A on the electrode surface can be used to follow the peroxidase catalyzed reaction [8,30,33,35–37]. In our work, the HRP is monitored by using hydrogen peroxide as enzyme substrate

and iodide as mediator (AH). So Reactions 2–4 can be summarized as:



The peroxidase enzyme of the secondary antibody (Ab* on Figure 1B) reacts with H_2O_2 (substrate) forming Compound I and water. This compound reacts with I^- (mediator) in two consecutive steps regenerating the enzyme and thus forming I_2 . This I_2 is probably present as I_3^- in the solution, which is reduced at the applied potential in the immunoassay analysis.

Figure 5 shows the LSV of the CDtrode modified with antigen (Ag) with a dilution of 1:10 and 0.5% BSA (blank) in 0.1 mol L^{-1} phosphate buffer solution at pH 7.0 containing $1 \times 10^{-4}/3 \times 10^{-4}$ mol L^{-1} $\text{H}_2\text{O}_2/\text{KI}$ (a) and CDtrodes modified with 1:10 Ag, 1:40 primary antibody (Ab) and 1:1000 secondary antibody conjugated with HRP enzyme (Ab*) in phosphate buffer solution (b), buffer solution with the substrate (c) and with substrate/mediator (d). For the next step -0.400 V was chosen to monitor the enzymatic reaction, because this potential presented a significant difference between the blank and the complete immunoassay.

3.4 Immunosensor Optimization

The optimization of the immunosensor was performed as described in item 2.4. In a first experiment, different dilutions of Ag and Ab were used. The Ab* dilution was fixed at 1:1000 and the substrate and mediator concen-

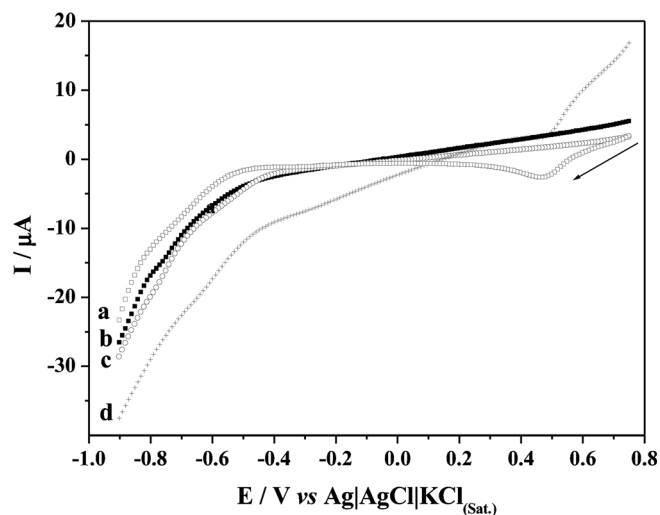


Fig. 5. Linear sweep voltammetry of Cdtrode modified with Ag and 0.5% BSA in 0.1 mol L^{-1} phosphate buffer solution pH 7.0 containing $\text{H}_2\text{O}_2/\text{KI}$ (a); CDtrode-Au modified with Ag, Ab and Ab* on 0.1 mol L^{-1} phosphate buffer solution at pH 7.0 (b), containing H_2O_2 (c) and $\text{H}_2\text{O}_2/\text{KI}$ (d); E/V vs $\text{Ag}|\text{AgCl}|\text{KCl}_{(\text{sat})}$; $v = 50 \text{ mV s}^{-1}$. The arrow indicates the direction of the sweep potential from the initial potential.

tration were 1×10^{-4} mol L^{-1} and 3×10^{-4} mol L^{-1} , respectively. Current intensity values obtained by chronoamperometric measurements (-0.400 V at 40 s) are presented in Figure 6; the Ag and Ab dilutions are shown as inset. The dilutions 1:20 of the antigen and 1:40 of the primary antibody corresponded to points within the linear range comprising the intersection of the curves. These dilutions were fixed for the next steps. In the intersection region of the curves, the current intensity is greater for a low variation in the concentration of Ag and Ab.

In the next step the Ab* dilution and H_2O_2 and KI concentration were investigated (inset in Figure 7). The methodology presented more sensibility at dilutions between 1:8000 and 1:4000. The Ab* dilution of 1:4000 and the H_2O_2 concentration of 1×10^{-4} and the KI con-

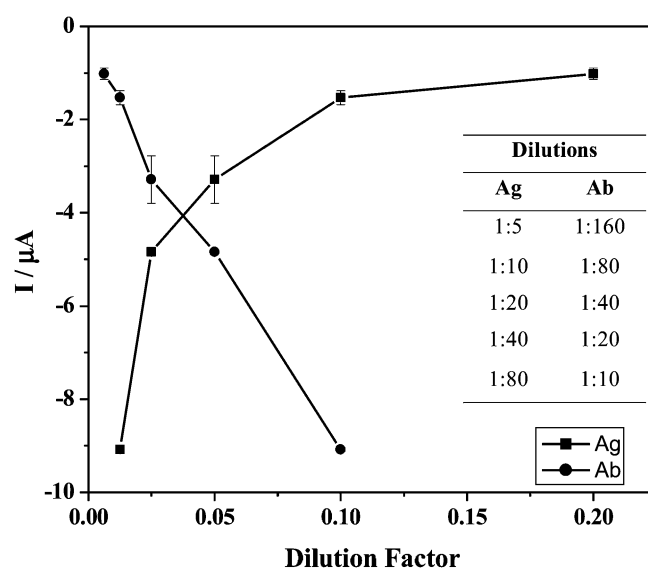


Fig. 6. Effect of antigen and primary antibody dilutions on the immunosensor response.

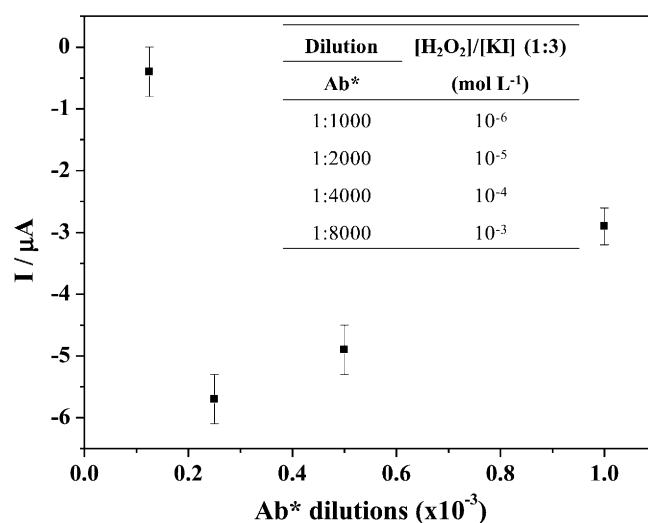


Fig. 7. Effects of secondary antibody tagged to peroxidase (HRP) dilution on the immunosensor response.

centration of 3×10^{-4} mol L⁻¹ were adopted for the development of the immunosensor.

3.5 Determination of the Cut-off on the Amperometric Immunoassay for Chagas' Disease

After the conditions for the immunosensor construction were defined, an investigation of the effect of negative sera was carried out. To determine the cut-off, ten sera from negative patients for Chagas' disease, previously analyzed by conventional methods, were employed in the developed immunosensor, and the average of these amperometric values was -0.409 ± 0.180 μ A. Three times the standard deviation was added to that average establishing finally the reactivity threshold of -0.949 μ A. Therefore, a sample was considered negative for Chagas' disease when the current intensity was equal or lower than -0.949 μ A, and positive when higher current values were measured.

3.6 Application of Amperometric Immunosensor to Chagasic Patients

The amperometric immunosensor was applied in four sera from positive patients for Chagas' disease, previously analyzed through conventional techniques (Table 1). In all the cases, the current intensity presented values more negatives than -0.949 μ A, leading to the conclusion that the serum of these patients containing antibodies for Chagas' disease.

Table 1. Values of current intensity for positive sera.

Positive sera	Current intensity (μ A)
01	-8.3 ± 0.9
02	-2.8 ± 0.8
03	-3.20 ± 0.03
04	-2.1 ± 0.3

4 Conclusions

The formation of self-assembled monolayers of SBZA on gold CDtrodes was efficient in a short time (30 min) and allowed the immobilization of Tc85 protein. This methodology demonstrated faster and lower cost than the construction of the same immunosensor with screen printed electrode (SPE) modified with cysteamine and glutaraldehyde, since the modification steps are six times faster and the transducer is much cheaper, i.e., the price of CDtrode is approximately 6% of the SPE. The proposed amperometric immunosensor had a cut-off of -0.949 μ A and showed to be able to detect the antibody anti-*T. cruzi* in sera of chagasic patients.

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